RIG-CLOSED POINTS AND RELATIVE FINITENESS FOR ADIC MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof proves (1) implies (3) through the finite-length argument and the complete-ring finiteness criterion, (3) implies (2) through integrality and spectral surjectivity, and (2) implies (1) through the residue-field lemma plus the Nullstellensatz

Editorial note. The label lemma-rig-closed-point-relative names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- rig-closed primes of Noetherian adic rings and the description of B/q as a Noetherian local ring of dimension 1 with its adic topology
- adicity of the composite A -> B/q when varphi is adic
- finite length of B/(q + varphi(a)) and the finiteness criterion for complete local rings (Algebra, Lemma algebra-lemma-finite-over-complete-ring)
- integrality implies surjectivity on spectra (Algebra, Lemma algebra-lemma-integral-overring-surjective)
- the residue field of a rig-closed point is finitely generated as an algebra over the residue field of the image (lemma-rig-closed-point-relative-residue-field of the same chapter, with lemma-finite-type-finite-type-red)
- the Hilbert Nullstellensatz: a field finitely generated as an algebra over a subfield is a finite extension (Algebra, Lemma algebra-lemma-field-finite-type-over-domain)

Source reference. Original lemma 00FY: sources/stacks-chow/tags.txt, source lines 588–588. Prerequisite pointer invoked at proof line(s) 3375; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 031D: sources/stacks-chow/tags.txt, source lines 3980–3980. Prerequisite pointer invoked at proof line(s) 3360; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00GQ: sources/stacks-chow/tags.txt, source lines 616–616. Prerequisite pointer invoked at proof line(s) 3366; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0GC5: sources/stacks-restricted/pinned-original-source.tex, source lines 2738–2738. Prerequisite pointer invoked at proof line(s) 3372; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGA: sources/stacks-restricted/pinned-original-source.tex, source lines 3314–3314. Prerequisite pointer invoked at proof line(s) 3371; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$ which is adic and topologically of finite type. Let $\mathfrak{q} \subset B$ be rig-closed. Let $\mathfrak{p} = \varphi^{-1}(\mathfrak{q}) \subset A$. Let $\mathfrak{a} \subset A$ be the ideal of topologically nilpotent elements. The following are equivalent*

(1) *the residue field κ of $B/\mathfrak{q}$ is finite over $A/\mathfrak{a}$,*
(2) *$\mathfrak{p} \subset A$ is rig-closed,*
(3) *$A/\mathfrak{p} \subset B/\mathfrak{q}$ is a finite extension of rings.*

Proof. Assume (1). Recall that $B/\mathfrak{q}$ is a Noetherian local ring of dimension 1 whose topology is the adic topology coming from the maximal ideal. Since φ is adic, we see that $A \to B/\mathfrak{q}$ is adic. Hence $\varphi(\mathfrak{a})$ is a nonzero ideal in $B/\mathfrak{q}$. Hence $B/\mathfrak{q} + \varphi(\mathfrak{a})$ has finite length. Hence $B/\mathfrak{q} + \varphi(\mathfrak{a})$ is finite as an $A/\mathfrak{a}$-module by our assumption. Thus $B/\mathfrak{q}$ is finite over A by Algebra, Lemma 031D. Thus (3) holds.

Assume (3). Then $\mathrm{Spec}(B/\mathfrak{q}) \to \mathrm{Spec}(A/\mathfrak{p})$ is surjective by Algebra, Lemma 00GQ. This implies (2).

Assume (2). Denote κ' the residue field of $A/\mathfrak{p}$. By Lemma 0GGA (and Lemma 0GC5) the extension κ/κ' is finitely generated as an algebra. By the Hilbert Nullstellensatz (Algebra, Lemma 00FY) we see that κ/κ' is a finite extension. Hence we see that (1) holds. □